TROUBLESHOOTING TOOLS MASTERY

1. Use the ping command to check the connectivity between your computer andwww.spotify.com. Note the response times and explain what they indicate about your network connection.

Command: `ping www.spotify.com`

Response Times & Indicator: Typical response times range from 15 ms to 45 ms depending on geographic proximity to the nearest content delivery network (CDN) edge node.

- **Low & Stable Response Times (≤ 30 ms):** Indicates a healthy, low-latency connection with minimal packet congestion.
- **High Jitter / Timeouts (>150 ms):** Indicates network congestion, weak Wi-Fi signal strength, or packet loss along the routing path.

2. Your friend is unable to access www.zomato.com, but other websites work fine. Usenslookup to check the DNS resolution for www.zomato.com and describe what the output tells you.

Command: `nslookup www.zomato.com`

Output Description: The `nslookup` utility queries the configured DNS resolver to translate the human-readable domain name into its corresponding IP address records (A or AAAA records).

- **Successful Output:** Displays the designated server IP addresses (e.g., AWS Route 53 or Cloudflare edge IPs associated with Zomato). This confirms that local DNS resolution is functioning correctly.
- **Error Output (e.g., SERVFAIL, NXDOMAIN):** Indicates that the local DNS server cannot reach root nameservers or that the domain name is invalid, meaning the issue is isolated to DNS resolution rather than general internet connectivity.

3. Open Wireshark and capture network traffic while you refresh your Instagram feed in abrowser. Identify and list the different protocols you see in the capture.

Capture Scenario: Capturing traffic while refreshing the Instagram feed.

Observed Protocols:

- **DNS:** Resolves API and media CDN hostnames (e.g., *.fbcdn.net).
- **TLSv1.2 / TLSv1.3:** Secures application layer data streams during encrypted HTTPS handshakes and session transfers.
- **TCP:** Manages reliable connection establishment, flow control, and packet retransmission.
- **HTTP/2 or HTTP/3 (QUIC):** Handles rapid, multiplexed retrieval of post imagery, videos, and JSON payload feeds.

4. A user reports that their Flipkart app is slow to load product images. Use netstat to list allcurrent network connections on your system and identify if there are any suspicious or high-usage connections that could be affecting bandwidth.

Command: `netstat -ano` (Windows) or `ss -tulnp` (Linux)

Analysis: Running `netstat` lists all active TCP and UDP socket connections along with their owning Process IDs (PIDs). To identify bandwidth hogs affecting the Flipkart app, look for established connections utilizing high port volumes or sustained high byte transmission counters. Cross-referencing suspicious PIDs in Task Manager helps isolate background updaters, cloud sync tools, or peer-to-peer applications consuming local bandwidth.

5. Your college WiFi keeps disconnecting during IPL match streaming. Create atroubleshooting checklist using ping, tracert, and netstat to systematically diagnose the issue step by step. For each tool, write what command you would use and what result would indicate a problem

Systematic Wi-Fi Troubleshooting Checklist for IPL Streaming:

- **Step 1: Local Gateway & Physical Link Check**
  Command: `ping 192.168.1.1` (or default gateway IP)
  Problem Indicator: High latency or "Request timed out" indicates local wireless interference, bad AP association, or a failing physical switch/router link.
- **Step 2: External Path & ISP Routing Check**
  Command: `tracert 8.8.8.8` (Windows) or `traceroute 8.8.8.8` (Linux)
  Problem Indicator: Packet loss or high latency appearing on the very first or second hop points to gateway congestion or overloaded campus core routers. If latency spikes further down the ISP path, the bottleneck is upstream with the internet service provider.
- **Step 3: Socket & Port Congestion Check**
  Command: `netstat -b` or `netstat -an`
  Problem Indicator: A saturated connection table or excessive sockets stuck in TIME_WAIT / CLOSE_WAIT states indicates application-level resource exhaustion or localized bandwidth throttling on the student network.